Drag, and why density is the limiting number

Why is a ten per cent density error worth an instrument program?

`Notebook companion: Unit-01-why-density.ipynb`

1 The question

NASA's Orbital Clarity Challenge asks for major-species density and temperature to ten per cent accuracy and ten per cent precision, over 90 to 300 km, with 10 km vertical and 5° horizontal resolution. Those numbers arrive as a specification. A specification given without the reasoning behind it is hard to argue with, and it is also hard to judge: the reader cannot tell which requirements are demanding and which are routine.

This unit derives what the specification is worth. It asks three questions. What does a density error do to a satellite operator? How large is the resulting effect, expressed in units an operator already uses? And how does ten per cent compare with the accuracy models deliver today, and with the tolerance of the decisions those models feed? Everything after this unit is about how to make the measurement. This one is about why anyone should want it.

2 Physical picture

A satellite at 400 km is in a near-vacuum, but not a perfect one, and the residual gas takes energy out of the orbit. The drag acceleration is

$$a_{\mathrm{drag}} \;=\; -\tfrac{1}{2}\,\rho\, v^2\,\frac{C_D A}{m} \;=\; -\frac{\rho v^2}{2B}, \qquad B \equiv \frac{m}{C_D A},$$

where ρ is the mass density of the local atmosphere and B is the ballistic coefficient. The ballistic coefficient collects the three spacecraft properties that matter into one number: mass, cross-sectional area and drag coefficient. A heavy spacecraft presenting a small area has a large B, and at a given density it experiences a small drag acceleration. A light spacecraft presenting a large area has a small B and decays faster. A compact satellite with a drag coefficient of 2.2 and an area-to-mass ratio of $0.01\ \mathrm{m^2/kg}$ has $B \simeq 45\ \mathrm{kg\,m^{-2}}$, and that satellite is the reference case used throughout this unit.

Everything in that expression is known to the operator except ρ. The mass and area are measured on the ground, the velocity comes from the orbit, and the drag coefficient is uncertain but bounded. The density is the term that is neither measured nor well modelled, and it varies by much more than the others.

The variation of the density is the reason this is an instrument problem. Between quiet conditions and a severe geomagnetic storm, the density at 400 km rises by a factor of 4.9, and at 600 km by a factor of 19.3 (section 4a). A storm is therefore the condition in which orbit prediction is hardest. It is also the condition in which an empirical model is least reliable, because such a model is fitted to a historical record in which quiet days greatly outnumber disturbed ones.

3 Derivation: why a density error compounds

A first guess would be that a ten per cent density error produces a ten per cent error in the predicted position, and that the error then stays that size. It does not. The error grows as the square of the elapsed time. The derivation is short and worth following, because that quadratic growth is what makes density accuracy valuable.

For a near-circular orbit, drag removes energy at a rate that shrinks the semi-major axis:

$$\frac{da}{dt} \;=\; -\frac{\rho\sqrt{\mu a}}{B}.$$

Suppose the density used for the prediction is wrong by a fraction f. The decay rate is proportional to the density, so it is wrong by that same fraction f. Integrating a constant rate error over a time t gives an error in semi-major axis that grows linearly with time, $\Delta a = f\,\dot{a}\,t$.

That error on its own would be harmless. A satellite a few metres lower than predicted is still close to where it was predicted to be, and no operator would care.

The consequence that matters is the second one. A satellite in a smaller orbit moves *faster*. Differentiating the mean motion $n = \sqrt{\mu/a^3}$ with respect to a gives $dn/da = -\frac{3}{2}\,n/a$, so an error in semi-major axis produces an error in angular rate. An error in angular rate, integrated over time, produces an error in position along the orbit. Integrating once more,

$$\Delta s \;=\; \tfrac{3}{4}\,n\,|\dot{a}|\,f\,t^2.$$

The quadratic growth is the point of the derivation. A density error is not a fixed offset that a margin can absorb once and for all. It is an error in the decay rate, which becomes an error in the angular rate, which then accumulates as an along-track position error for as long as the prediction runs. Doubling the prediction horizon therefore quadruples the position error.

The derivation held the density, the decay rate and the mean motion fixed. A real orbit decays into denser air, so all three of them change slowly as the satellite descends. Section 4b checks the closed form against a numerical integration that lets them change, and the two agree to two and a half per cent over a week. That is close enough for the arguments made here.

4 Worked numbers

4a. The atmosphere the satellite is flying through

Altitude	Quiet	Storm, ap = 400	Ratio
200 km	2.77×10^{-10}	4.64×10^{-10}	1.7
300 km	2.03×10^{-11}	5.81×10^{-11}	2.9
400 km	2.41×10^{-12}	1.18×10^{-11}	4.9
500 km	3.51×10^{-13}	3.24×10^{-12}	9.2
600 km	6.42×10^{-14}	1.24×10^{-12}	19.3

Total neutral mass density in kg m^{-3} from NRLMSIS 2.1 [3], including the helium that the instrument's optical model omits.

The last column is the one to carry forward. A storm raises the density by a factor of 1.7 at 200 km and by a factor of 19.3 at 600 km. The storm-to-quiet ratio grows with altitude because a storm heats the thermosphere, heating increases the scale height, and increasing the scale height raises the density high in the profile far more than it does near the bottom. A model asked to predict during a storm at high altitude is therefore working where its record is thinnest and where the departure from the historical average is largest.

4b. What a ten per cent error costs

Altitude	State	Decay rate	1 day	3 days	7 days
300 km	quiet	1989 m/day	14.9 km	134 km	732 km
300 km	storm	5692 m/day	42.7 km	385 km	2094 km
400 km	quiet	238 m/day	1.75 km	15.7 km	85.6 km
400 km	storm	1160 m/day	8.52 km	76.7 km	417 km
500 km	quiet	35 m/day	0.25 km	2.26 km	12.3 km
500 km	storm	322 m/day	2.31 km	20.8 km	113 km

Along-track position error accumulated from a ten per cent density error, for a satellite with $B = 45\,\mathrm{kg\,m^{-2}}$.

Against a proper numerical integration the closed form gives 1.75 against 1.75 km at one day, 15.72 against 15.89 at three days and 85.6 against 87.7 at seven. The approximation is good to two and a half per cent over a week and it errs low, because the real orbit decays into denser air.

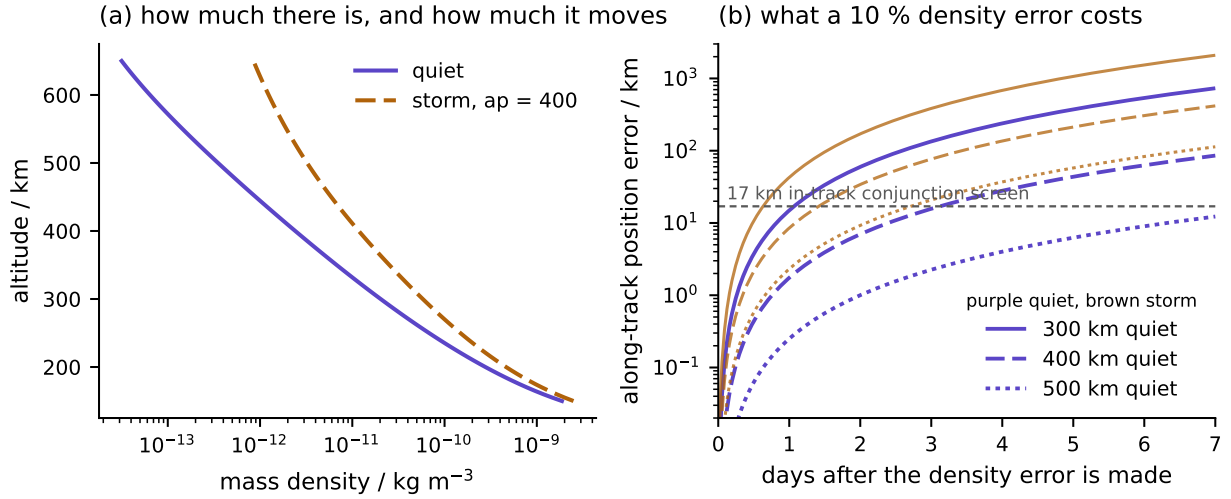


Figure 1: (a) Total neutral mass density against altitude, quiet and severely disturbed. The gap widens with altitude. (b) Along-track position error accumulated from a ten per cent density error, at three altitudes, with the in-track dimension of NASA's conjunction screening box marked. Generated by section D of the notebook.

4c. Ten per cent against the alternatives

Two comparisons are needed before the table above becomes an argument. The first is with the accuracy that models deliver now. The second is with the tolerance of the decisions that use the predictions.

The first comparison is with current models. The paper describing the High Accuracy Satellite Drag Model states plainly that "current atmospheric density models have errors of 15–20 %" [6], and that model was built to improve on them. Independently, the SUVI occultation retrievals found NRLMSIS wrong by −26 % at dawn [5]. Ten per cent is not an arbitrary target; it is roughly half of the current error.

The second comparison is with the tolerance of the decisions that use the prediction. NASA's Conjunction Assessment Risk Analysis screens close approaches inside a box of 0.5 km radial by 17 km in-track by 20 km cross-track, looking seven days ahead [4]. A density error displaces a satellite along its own track rather than across it, so the relevant dimension of that box is the in-track one, and it is 17 km.

Density error	Quiet	Storm
10 %, the benchmark	3.12 days	1.41 days
17.5 %, today's models	2.36 days	1.07 days
30 %, a disturbed day	1.80 days	0.82 days

Time for a density error alone to displace a satellite at 400 km by 17 km along track, which is the in-track dimension of the screening box.

Read those figures as an operator would. With today's models, the predicted position of a satellite at 400 km differs from its true position by more than the 17 km width of its own screening box after about two and a half days in quiet conditions, and after about one day during a storm, from the density error alone. Reducing the density error to ten per cent extends those intervals to 3.12 days and 1.41 days: roughly three quarters of a day of extra prediction in quiet conditions, and about a third of a day during a storm. That gain is the measurable value of the instrument, and it is the answer to give a reviewer who asks why ten per cent is the number.

4d. Why the measurement is scarce

Accuracy is one of the two things a measurement programme can improve. The other is sampling: how many profiles are measured, and at what altitudes, local times and geomagnetic conditions. Sampling is the one that a cheap instrument changes.

A density instrument today costs between one and ten million dollars. That price is not arbitrary. It pays for absolute radiometric calibration, mechanisms, high-voltage supplies, and the qualification programme each of those requires. The consequence is that very few fly, and the empirical models everyone uses are fitted to a thin and unevenly distributed record — largely accelerometer-derived densities from a handful of research satellites, supplemented by orbit-decay inferences from tracked objects [1]. A model fitted to that record has to interpolate across large gaps in altitude, local time and geomagnetic activity. The gaps are widest for disturbed conditions, because storms are

infrequent and the satellites carrying accelerometers were few. The storm-time behaviour shown in section 4a is therefore the part of the model supported by the least data.

An occultation instrument produces two profiles per orbit, one at each terminator crossing, when the beta angle permits (Unit 5, section 4c). At 500 km that is 15.2 orbits per day, so 30.5 profiles per day from a single unit and about 11 100 in a year. A hundred hosted units would return roughly 3000 profiles a day.

That is the relevance argument, and it is a sampling argument rather than an accuracy one. A single instrument ten per cent better than the state of the art is a useful contribution. A hundred instruments each ten per cent better, distributed across altitudes and inclinations and returning profiles continuously, is a different kind of contribution, and it is the one the hosted-payload model makes possible. Every design decision in the rest of this series that gives up capability in exchange for lower cost is made in order to keep a hundred units affordable.

4e. The benchmark, read as a specification

Each row of the benchmark is limited by a different part of the design, and knowing which is which determines where effort belongs. The standing shown here is what Unit 6 and Unit 8 establish; it is repeated at the front of the series so that the rest of it can be read as an argument rather than a tour.

Benchmark row	Limited by	Where it is settled
Accuracy, 10 %	cross sections	Unit 4. A literature floor, not a hardware one.
Precision, 10 %	detector stability	Unit 6. Flat-field drift sets the usable window.
Altitude, 90–300 km	channel set	Unit 6. Per species, and short at the bottom.
Vertical, 10 km	the inversion	Unit 7. Sampling, not resolution.
Horizontal, 5°	geometry	Unit 5. Intrinsic to a limb path.
Temperature	the inversion	Unit 7. Derived, not measured.

Only one of those rows is set by hardware we buy. That is the shape of the whole project: building the instrument is the easier part, and inverting its measurements into densities and temperatures is the harder one. We state that with some confidence because five days of simulation in early September were run to test the opposite proposition, and the results did not support it.

5 What the computation shows

A density error compounds with time, so density accuracy is better expressed as days of prediction than as kilometres of position. At 400 km in quiet conditions, a ten per cent density error takes 3.12 days to displace a satellite by the 17 km in-track dimension of the conjunction screening box, against 2.36 days at the 17.5 per cent error current models carry.

The difference between quiet and disturbed conditions grows with altitude. A storm-to-quiet density ratio of 1.7 at 200 km becomes 19.3 at 600 km. An instrument that works only in quiet conditions therefore addresses the easier half of the problem, which is why the design case in Unit 3 is a storm.

Most of the benchmark is not about the hardware. Five of the six rows are set by cross sections, geometry or the inversion. That is worth knowing before reading nine more units about filters and detectors.

6 What we decided

The submission leads with days of prediction rather than per cent of density. The benchmark is stated in per cent because that is what an instrument delivers, but the value to an operator is the additional prediction time it provides, and section 4c is the conversion between the two.

We claim improvement against a stated baseline rather than absolute accuracy. Ten per cent against the 15 to 20 per cent that current models carry is a defensible claim with a citation behind it; "ten per cent accuracy" on its own invites the question of what that is better than.

We do not propose to meet every benchmark row. Unit 5 and Unit 6 show which rows are intrinsic to a limb measurement and which are short, and NASA states explicitly that an applicant may propose a different target with justification. Section 4e is where the justification is organised.

7 What we do not claim

Not that the ballistic coefficient is known. Every number in section 4b assumes $B = 45\,\mathrm{kg\,m^{-2}}$ exactly. In practice the drag coefficient of a real satellite in free molecular flow is uncertain at the ten per cent level and depends on attitude, surface composition and gas–surface accommodation. That uncertainty enters the along-track error in exactly the same way a density error does, and no atmospheric instrument reduces it. The two are separable in principle, because the density varies with geomagnetic activity and the drag coefficient does not.

Not that our instrument measures the density that drag responds to. Drag responds to total mass density along the orbit. We measure species columns along a limb path at the terminator (Unit 5), from which a mass density has to be constructed. The two quantities are closely related but they are not the same quantity, and making the conversion is the responsibility of whoever assimilates the product.

Not that the conjunction figures are a performance claim. The screening box is a NASA operational parameter from an agency technical paper [4] rather than a peer-reviewed result, and the time-to-cross figures assume the density error is the only error in the prediction. It is not. They are a way of putting the benchmark into units an operator recognises, not a prediction of operational benefit.

Not that the quadratic growth law holds indefinitely. It comes from a linearisation about a circular orbit with a fixed decay rate. Over weeks, or for an orbit low enough to be decaying quickly, the satellite descends far enough that it meets denser air, the decay rate itself increases, and the position error grows faster than quadratically. Section 4b shows the closed form already running two and a half per cent low at seven days.

8 Exercises

E1.1 The along-track error grows as $f\,t^2$, so the time to accumulate a fixed error scales as $f^{-1/2}$. Use that to find the density accuracy needed to hold the along-track error below one kilometre for three days at 400 km. Compare your answer with the ten per cent benchmark, and say what the comparison tells you about what the benchmark is and is not for.

E1.2 Repeat the table of section 4b for a satellite whose area-to-mass ratio is ten times larger, which is roughly a CubeSat with deployed panels. Find the altitude at which a ten per cent density error costs that satellite as much position error in one day as our reference satellite accumulates in seven.

E1.3 The storm-to-quiet density ratio in section 4a grows with altitude. Explain why in terms of scale height, using Unit 2 if you have read it. Then decide which is worse for an operator at 600 km: a model that is twenty per cent wrong all the time, or one that is five per cent wrong when quiet and wrong by a factor of two during storms. State the assumption about storm frequency that your answer depends on.

References

[1] J. T. Emmert, "Thermospheric mass density: a review," *Adv. Space Res.* **56**(5), 773–824 (2015), doi:10.1016/j.asr.2015.05.038.

[2] J. T. Emmert, H. P. Warren, A. M. Segerman, J. M. Byers and J. M. Picone, "Propagation of atmospheric density errors to satellite orbits," *Adv. Space Res.* **59**(1), 147–165 (2017), doi:10.1016/j.asr.2016.07.036.

[3] J. T. Emmert, D. P. Drob, J. M. Picone, D. E. Siskind, M. Jones Jr. et al., "NRLMSIS 2.0: a whole-atmosphere empirical model of temperature and neutral species densities," *Earth Space Sci.* **8**(3), e2020EA001321 (2021), doi:10.1029/2020EA001321.

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[5] R. K. Sewell et al., "Thermospheric composition and temperature from GOES-R/SUVI solar occultations," *Space Weather* **23**, e2024SW004234 (2025); arXiv:2508.10242.

[6] M. F. Storz, B. R. Bowman, J. I. Branson, S. J. Casali and W. K. Tobiska, "High accuracy satellite drag model (HASDM)," *Adv. Space Res.* **36**(12), 2497–2505 (2005), doi:10.1016/j.asr.2004.02.020.

Every quantitative result in this unit is provisional in the four ways that apply to the whole project: filter bands are idealized top-hats rather than measured transmission curves; the truth atmosphere is NRLMSIS 2.1; helium is omitted above 400 km; and no horizontal structure along the ray is modelled. Analysis code is in `channel-plan/`.